LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Work and Energy

Chapter / Topic: Work and Energy (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Take g = 10 m/s^2 unless stated otherwise.

SECTION A

6 marks

Choose the correct option.

Q1. The SI unit of work is the: **[1]**

(a) newton (b) watt
(c) joule (d) pascal

Q2. 1 kWh is equal to: **[1]**

(a) 3.6×10^6 J (b) 3.6×10^3 J
(c) 1000 J (d) 3600 J

Q3. When an object is raised to a height, which form of its energy increases? **[1]**

(a) Kinetic energy (b) Heat energy
(c) Sound energy (d) Potential energy

Q4. If the velocity of a body is doubled, its kinetic energy becomes: **[1]**

(a) 2 times (b) 4 times
(c) half (d) unchanged

Q5. The work done by a force acting at right angles to the displacement is: **[1]**

(a) maximum (b) zero
(c) negative (d) infinite

Q6. **Assertion (A):** A man standing still with a heavy suitcase in his hand does no work on it (in the scientific sense).
Reason (R): Work is done only when a force produces a displacement in its direction. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Define 1 joule. How much work is done in lifting a 2 kg bag to a height of 1.5 m? **[2]**

OR

What is the difference between energy and power?

Q8. A 60 W bulb is used for 5 hours every day. How much energy does it use in 30 days, in kWh? **[2]**

SECTION C

6 marks

Q9. A 2 kg ball is dropped from a height of 5 m. (Ignore air resistance.) **[3]**

(i) Find its potential energy at the top. **[1]**

(ii) Find its kinetic energy just before it hits the ground. **[1]**

(iii) Find its speed just before it hits the ground. **[1]**

Q10. Describe the energy changes in (i) a hydroelectric power plant, (ii) a battery-powered torch, (iii) a swinging pendulum. **[3]**

OR

Show that the kinetic energy of an object of mass m moving with velocity v is $\frac{1}{2}mv^2$.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is power? Give its SI unit and define 1 watt. **[2]**

(ii) A girl of mass 50 kg climbs 20 steps, each 20 cm high, in 10 s. Find the work she does and her power. **[3]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A pendulum bob of mass 0.2 kg is pulled aside so that it is 0.2 m above its lowest point, and then released. It swings back and forth, but after some time it stops. **[4]**

(i) Find its potential energy at the highest point (relative to the lowest point). **[1]**

(ii) What is its kinetic energy at the lowest point (ignoring friction)? **[1]**

(iii) Find its speed at the lowest point. Why does it finally stop? Is energy destroyed? **[2]**

~ All the best! ~